



- *Data (0 or more octets)*

The Data field is variable in length. The Code field determines the format of the Data field.

If a device does not support pairing then it shall respond with a Pairing Failed command with the reason set to “Pairing Not Supported” (see [Section 3.5.5](#)) when any command is received. If pairing is supported then all commands shall be supported.

3.4 SMP TIMEOUT

To protect the Security Manager protocol from stalling, a Security Manager Timer is used. Upon transmission of the Security Request command or reception of the Security Request command, the Security Manager Timer shall be reset and restarted. Upon transmission of the Pairing Request command or reception of the Pairing Request command, the Security Manager Timer shall be reset and started.

The Security Manager Timer shall be reset when an L2CAP SMP command is queued for transmission.

When a Pairing process completes, the Security Manager Timer shall be stopped.

If the Security Manager Timer reaches 30 seconds, the procedure shall be considered to have failed, and the local higher layer shall be notified. No further SMP commands shall be sent over the L2CAP Security Manager Channel. A new Pairing process shall only be performed when a new physical link has been established.

3.5 PAIRING METHODS

The SMP commands defined in this section are used to perform Pairing Feature Exchange and key generation (see [Section 2.1](#)).

3.5.1 Pairing Request

The initiator starts the Pairing Feature Exchange by sending a Pairing Request command to the responding device. The Pairing Request command is defined in [Figure 3.2](#).

The rules for handing a collision between a pairing procedure on the LE transport and a pairing procedure on the BR/EDR transport are defined in [\[Vol 3\] Part C, Section 14.2](#).

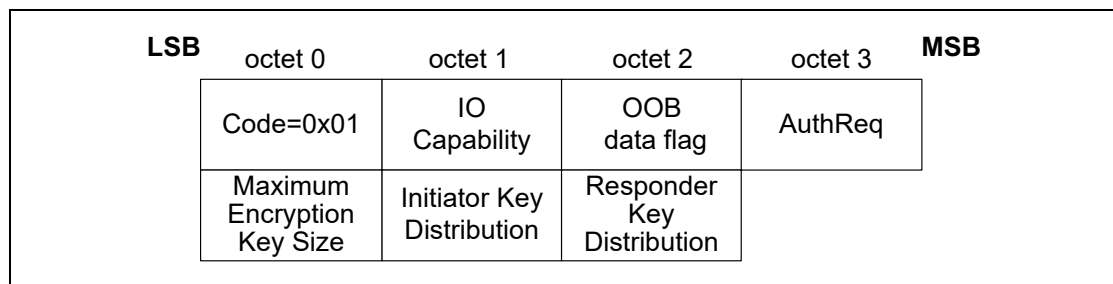


Figure 3.2: Pairing Request Packet

The following data fields are used:

- *IO Capability (1 octet)*

[Table 3.4](#) defines the values which are used when exchanging IO capabilities (see [Section 2.3.2](#)).

Value	Description
0x00	DisplayOnly
0x01	DisplayYesNo
0x02	KeyboardOnly
0x03	NoInputNoOutput
0x04	KeyboardDisplay
0x05-0xFF	Reserved for future use

Table 3.4: IO Capability Values

- *OOB data flag (1 octet)*

[Table 3.5](#) defines the values which are used when indicating whether OOB authentication data is available (see [Section 2.3.3](#)).

Value	Description
0x00	OOB Authentication data not present
0x01	OOB Authentication data from remote device present
0x02-0xFF	Reserved for future use

Table 3.5: OOB Data Present Values

- *AuthReq (1 octet)*

The AuthReq field is a bit field that indicates the requested security properties (see [Section 2.3.1](#)) for the STK and LTK and GAP bonding information (see [\[Vol 3\] Part C, Section 9.4](#)).

[Figure 3.3](#) defines the authentication requirements bit field.

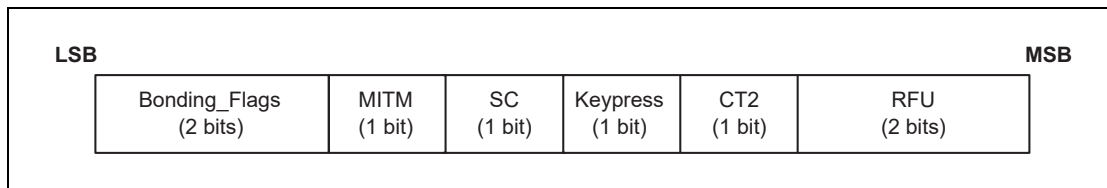


Figure 3.3: Authentication Requirements Flags

The Bonding_Flags field is a 2-bit field that indicates the type of bonding being requested by the initiating device as defined in [Table 3.6](#).

Bonding_Flags b_1b_0	Bonding Type
00	No Bonding
01	Bonding
10	Reserved for future use
11	Reserved for future use

Table 3.6: Bonding Flags

The MITM field is a 1-bit flag that is set to one if the device is requesting MITM protection, otherwise it shall be set to 0. A device sets the MITM flag to one to request an Authenticated security property for the STK when using LE legacy pairing and the LTK when using LE Secure Connections.

The SC field is a 1 bit flag. If LE Secure Connections pairing is supported by the device, then the SC field shall be set to 1, otherwise it shall be set to 0. If both devices support LE Secure Connections pairing, then LE Secure Connections pairing shall be used, otherwise LE Legacy pairing shall be used.

The keypress field is a 1-bit flag that is used only in the Passkey Entry protocol and is ignored in other protocols. When both sides set that field to one, Keypress notifications shall be generated and sent using SMP Pairing Keypress Notification PDUs.

The CT2 field is a 1-bit flag that shall be set to 1 upon transmission to indicate support for the *h7* function. See sections [2.4.2.4](#) and [2.4.2.5](#).

- *Maximum Encryption Key Size (1 octet)*

This value defines the maximum encryption key size in octets that the device can support. The maximum key size shall be in the range 7 to 16 octets.

- *Initiator Key Distribution / Generation (1 octet)*

The Initiator Key Distribution / Generation field indicates which keys the initiator is requesting to distribute / generate or use during the Transport Specific Key Distribution phase (see [Section 2.4.3](#)). The Initiator Key Distribution / Generation field format and usage is defined in [Section 3.6.1](#).

- *Responder Key Distribution / Generation (1 octet)*



The Responder Key Distribution / Generation field indicates which keys the initiator is requesting the responder to distribute / generate or use during the Transport Specific Key Distribution phase (see [Section 2.4.3](#)). The Responder Key Distribution / Generation field format and usage is defined in [Section 3.6.1](#).

If Secure Connections pairing has been initiated over BR/EDR, the following fields of the SM Pairing Request PDU are reserved for future use:

- the IO Capability field,
- the OOB data flag field, and
- all bits in the Auth Req field except the CT2 bit.

3.5.2 Pairing Response

This command is used by the responding device to complete the Pairing Feature Exchange after it has received a Pairing Request command from the initiating device, if the responding device allows pairing. The Pairing Response command is defined in [Figure 3.4](#).

The rules for handing a collision between a pairing procedure on the LE transport and a pairing procedure on the BR/EDR transport are defined in [\[Vol 3\] Part C, Section 14.2](#).

If a Pairing Request is received over the BR/EDR transport when either cross-transport key derivation/generation is not supported or the BR/EDR transport is not encrypted using a Link Key generated using P256, a Pairing Failed shall be sent with the error code "Cross-transport Key Derivation/Generation not allowed" (0x0E).

LSB	octet 0	octet 1	octet 2	octet 3	MSB
	Code=0x02	IO Capability	OOB data flag	AuthReq	
	Maximum Encryption Key Size	Initiator Key Distribution	Responder Key Distribution		

Figure 3.4: Pairing Response Packet

The following data fields are used:

- *IO Capability (1 octet)*
[Table 3.4](#) defines the values which are used when exchanging IO capabilities (see [Section 2.3.2](#)).
- *OOB data flag (1 octet)*
[Table 3.5](#) defines the values which are used when indicating whether OOB authentication data is available (see [Section 2.3.3](#)).